

Charotar University of Science and Technology
Education Campus, Changa - 388 421.

M Sc (BIOTECHNOLOGY/ MICROBIOLOGY)
SEMESTER II EXAMINATION
BT708/MI708: INDUSTRIAL BIOTECHNOLOGY

Date: 1st May, 2010
Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in Part A should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of Part B in separate answer sheets.

PART A

Total marks: 20

Q1. Answer as directed:

(1) For a biotechnology company to be listed in stock market it should have

- a. At least 10 products in clinical trials
- b. More than 30 products in clinical trials
- c. At least 2 products in clinical trials
- d. None of the above

(2) Approach taken by a biotechnology company for discovery of new drug is

- a. Top down
- b. Bottom up
- c. Top down and bottom up
- d. Top down or bottom up

(3) Which of the following is NOT a government supports given to biotechnology companies

- a. Technology transfer scheme
- b. Small company support scheme
- c. Company development scheme
- d. a, b and c

(4) FIPCO stands for _____

(5) Citrate serves as an inhibitor of phosphofructokinase.

☐ TRUE ☐ FALSE

(6) Following amino acid DOES NOT belong to aspartate family

- a. Lysine
- b. Methionine
- c. Threonine
- d. glutamine

(7) Which of the following structure is formed when Y group in the cobalamins is $-\text{CH}_3$

- a. Mecobalamin
- b. Vit B12 a
- c. Vit B12
- d. Coenzyme B12

(8) Under insufficient oxygen *Brevibacterium flavum* accumulates _____ in L- glutamic acid fermentation

(9) β -exotoxin produce by *B thuringiensis* is known as _____

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(10) Amylase belongs to _____ class of enzymes.

(11) A-8 auxotrophic mutant used in rifamycin production lacks

a. Transketolase

b. Shikimate phosphorylation

c. Both

d. None of the above

(12) GRAS stands for genetically regarded as safe.

☐ TRUE ☐ FALSE

(13) Which of the following organism is **NOT** used as probiotic.

a. Bifidobacteria

b. Lactobacilli

c. *F. Oxysporum*

d. a, b & c

(14) _____ proteins simplifies downstream process.

(15) The expression of recombinant proteins in cells in which they naturally occur is termed as

a. homologous

b. heterologous

c. extracellular

d. intracellular

(16) Good manufacturing practices are followed in industry.

☐ TRUE ☐ FALSE

(17) Chemotherapy drugs are listed in

(a) P category

(b) U category

(c) D category

(d) Z category

(18) EPA stands for environment pollution agency.

☐ TRUE ☐ FALSE

(19) Planned and systemic actions necessary to provide adequate confidence that a product or service will satisfy given requirements for quality is _____

(20). First step in validation is

(a) operational qualification

(b) installation qualification

(c) performance qualification

(d) none of the above

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2. Write answers for Section I and Section II of Part B in separate answer sheets.

Part B

Total Marks: 50

Section I

- Q1. (A) Discuss the types of government supports given to biotechnology companies. [5]
Q1. (B) Give an outline of clinical trials with importance of each stage. [5]

OR

- Q1. (A) Write a note on requirements of patentability and Parts of Patent Application. [5]
Q1. (B) What is containment? Describe the various types of containment. [4]
Q1. (C) Enlist requirements for a biotech companies to be listed in stock market. [1]

- Q2. (A) Give methods to prevent inclusion bodies formation during recombinant protein production [2]
Q2. (B) Describe various criteria for using a microorganism as a probiotic. [3]
Q2. (C) Give 3 examples of nutraceutical products with their use. [3]
Q2. (D) Enlist benefits of using transgenic plants for recombinant protein production. [2]

OR

- Q2. (A) Describe in detail various steps in recombinant protein production [5]
Q2. (B) Discuss: Types of Nutraceuticals [5]

Section II

- Q3. (A) Discuss: Citric acid fermentation process [5]
Q3. (B) Discuss: Vitamin B12 fermentation process [5]

OR

- Q3. (A) Give a schematic diagram of citric acid recovery [5]
Q3. (B) Give applications of (i) xanthan gum (ii) ethanol [3]
Q3. (C) Enlist the organisms used in production of glutamic acid and lysine [2]

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Q4. (A) Discuss: amylase production

[5]

Q4. (B) Discuss briefly the preparation of inoculant for nitrogen fixation

[5]

OR

Q4. (A) Discuss: Biosynthesis of rifampicin S

[5]

Q4. (B) Define Bioinoculants and give examples of organisms used as bioinoculants.

[3]

Q4. (C) Enlist applications of amylase

[2]

Q5. (A) Give a Flow diagram of manufacturing facility for biotechnology based products

[2]

Q5. (B) Discuss: validation in biotech industry

[3]

Q5. (C) Write a note on “Principles of Quality assurance”

[2]

Q5. (D) Write a note on: Biowaste treatment

[3]

OR

Q5. (A) Describe the types of water used in biotech companies

[5]

Q5. (B) Write the differences between GLP and GMP

[2]

Q5. (C) Enlist the variables of quality assurance and describe any one variable in detail

[3]